SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA04 **COURSE NAME: Operating Systems**

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.* *(BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Concept Mapping

Code of Conduct:

I Gowtham M(192373060) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student Gowtham

Annexure A

Concept Mapping

Question 1 **10 Marks**

Concept Mapping: Operating System Components and Services

An Operating System provides several services to users and application programs.

Task:

Match each Operating System Component with the most appropriate Operating System Service and explain the relationship.

Components	Services
Process Manager	File Management
Memory Manager	Process Execution
File Manager	Memory Allocation
Device Manager	I/O Operations
Security Manager	Protection & Security

After mapping, explain how these components collectively improve overall system performance.

Question 2 **10 Marks**

Concept Mapping: System Calls and Operating System Services

An application performs the following operations:

- Creates a new process
- Reads a file
- Allocates memory
- Sends data through a pipe
- Terminates execution

Task:

Map each operation to the appropriate system call category.

Operations	Categories
Create Process	Communication
Read File	File Management
Allocate Memory	Process Control
Send Data	Memory Management
Exit Process	Process Control

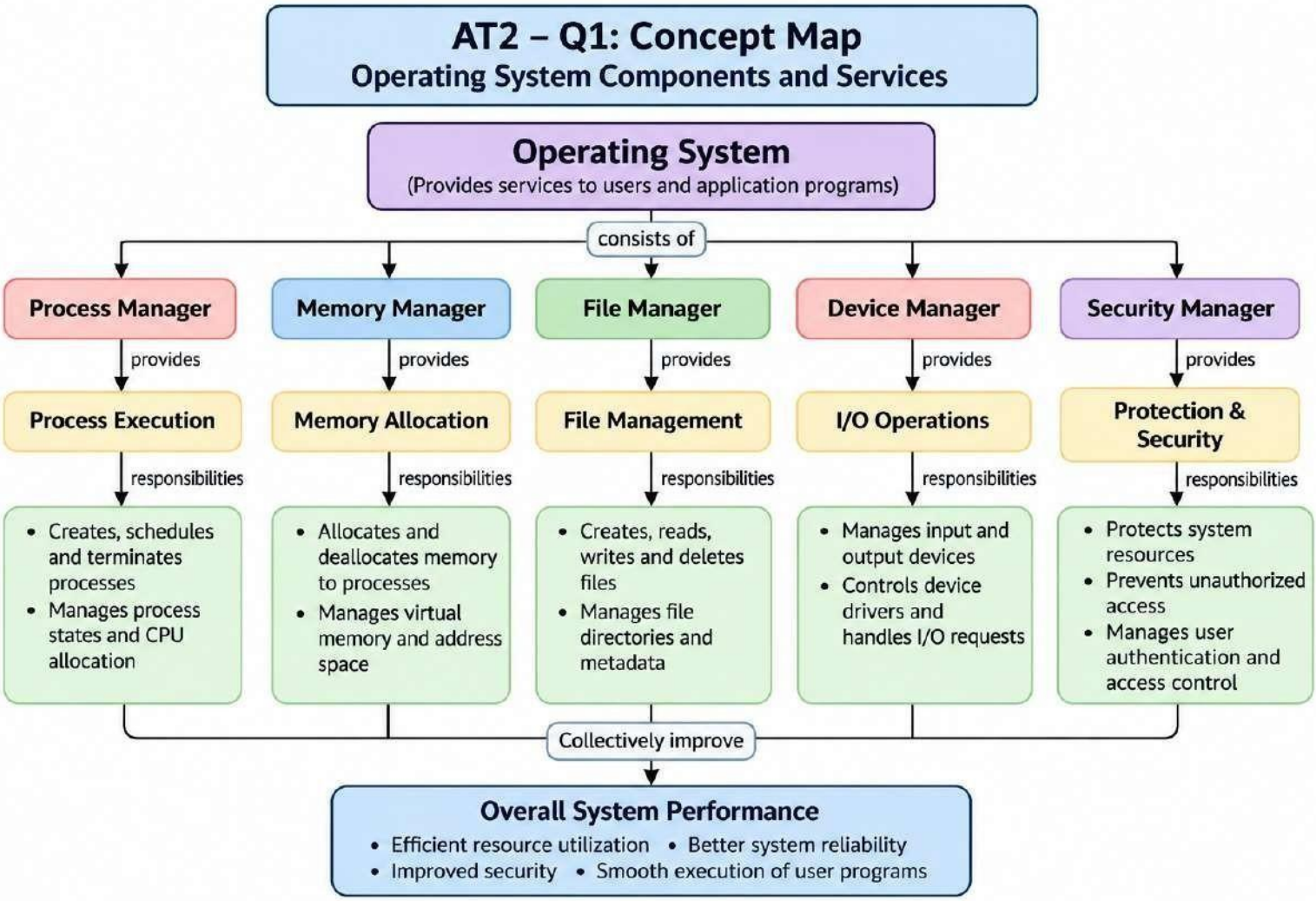
Write one example system call corresponding to each mapped category.

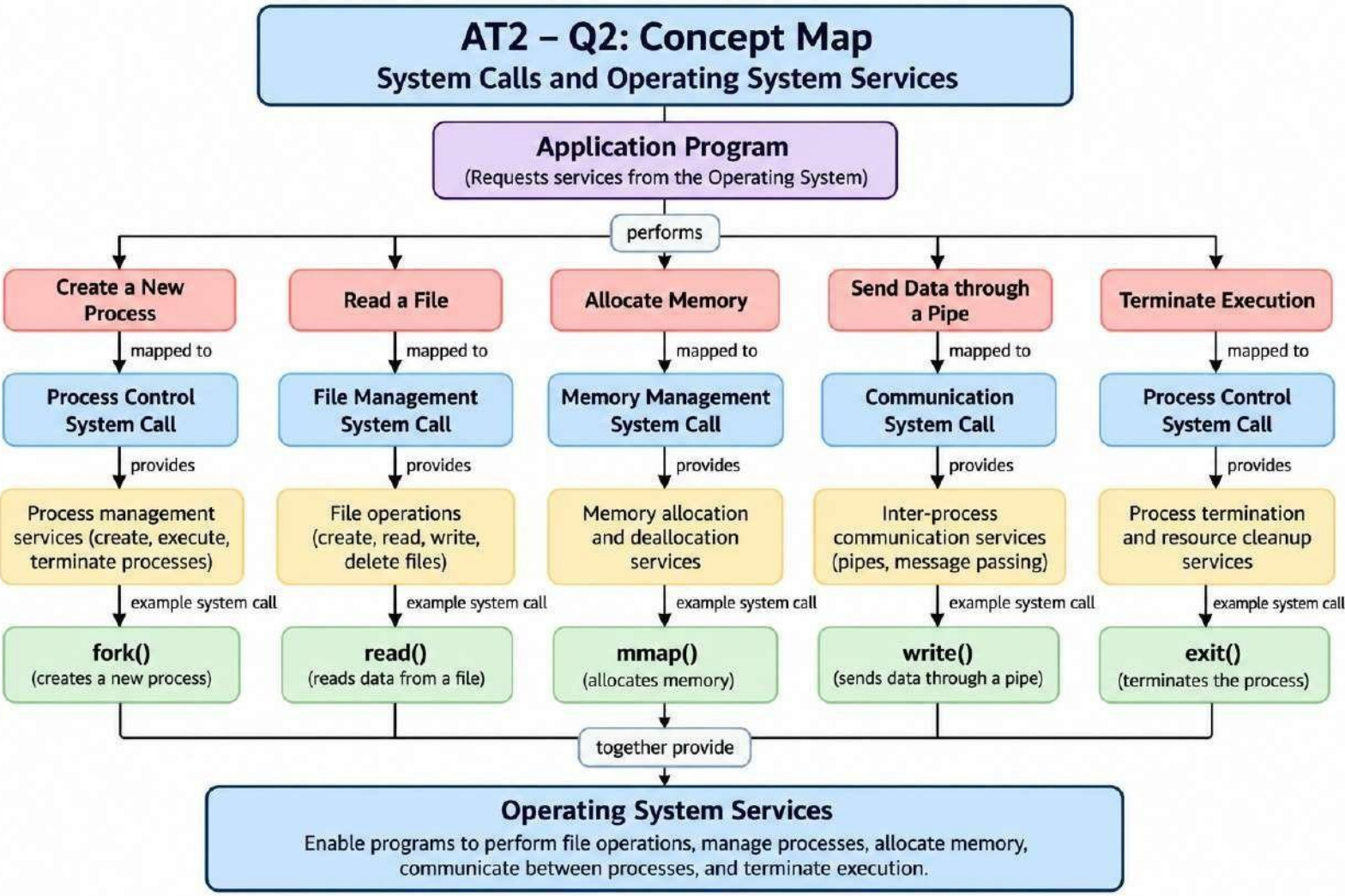
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are wellstructured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (100%)
Q1						
Q2						
Total						


AT2 – Q1: Concept Map
Operating System Components and Services
Operating System
(Provides services to users and application programs)
consists of
Process Manager
Memory Manager
File Manager
Device Manager
Security Manager
provides
Process Execution
Memory Allocation
File Management
I/O Operations
Protection & Security
responsibilities
• Creates, schedules and terminates processes
• Manages process states and CPU allocation
• Allocates and deallocates memory to processes
• Manages virtual memory and address space
• Creates, reads, writes and deletes files
• Manages file directories and metadata
• Manages input and output devices
• Controls device drivers and handles I/O requests
• Protects system resources
• Prevents unauthorized access
• Manages user authentication and access control
Collectively improve
Overall System Performance
• Efficient resource utilization • Better system reliability
• Improved security • Smooth execution of user programs


AT2 – Q2: Concept Map
System Calls and Operating System Services
Application Program
(Requests services from the Operating System)
performs
Create a New Process
Read a File
Allocate Memory
Send Data through a Pipe
Terminate Execution
mapped to
Process Control System Call
File Management System Call
Memory Management System Call
Communication System Call
Process Control System Call
provides
Process management services (create, execute, terminate processes)
File operations (create, read, write, delete files)
Memory allocation and deallocation services
Inter-process communication services (pipes, message passing)
Process termination and resource cleanup services
example system call
fork()
(creates a new process)
read()
(reads data from a file)
mmap()
(allocates memory)
write()
(sends data through a pipe)
exit()
(terminates the process)
together provide
Operating System Services
Enable programs to perform file operations, manage processes, allocate memory, communicate between processes, and terminate execution.